

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							Semester: I		
Course: Electronics Circuit Design							Code: ECES102		
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
02	02	-	03	40	60	50	25	-	175
Prerequisites:									
Basics of physics									
Course Objectives:									
1. To enable students to comprehend and identify various electronic symbols and components used in electrical and electronic circuits.									
2. To develop the ability to design and analyze basic electronic circuits involving resistors, capacitors, inductors, and special purpose diodes.									
3. To equip students with the skills to apply PCB design guidelines and construct oscillator and amplifier circuits effectively.									
Course Outcomes: After completion of this course, students will able to -									
CO1	Identify and interpret the symbols for various electronic components, demonstrating foundational knowledge.								
CO2	Calculate the equivalent capacitance, resistance, and inductance in series and parallel circuits.								
CO3	Design and analyze circuits using special purpose diodes, including Zener and tunnel diodes.								
CO4	Test and troubleshoot electronic components and circuits using millimeters and oscilloscopes.								
CO5	Develop PCB layouts following standard guidelines for single-sided and double-sided boards.								
CO6	Design, construct, and analyze various types of oscillators and amplifiers, applying theoretical concepts to practical implementations.								
Course Contents:									
Unit	Description								Duration (Hrs.)
1.	Introduction to electronics Symbols: Series and parallel connections of resistors, capacitor, and inductors, Calculation of equivalent capacitance resistance and inductance, symbols of LDR, potentiometer, 555, toggle switch. USB, Interface and Connector, Reset Button, Power button, Digital Input/output pins, Analog Pins, Microprocessor and Microcontroller block diagram								5
2.	Introduction electrical component and Symbol: D.C., A.C., Positive, Negative, Single Phase A.C. 50 Hz , Three Phase A.C., 50 Hz, A.C. / D.C., 3-Phase line, Neutral line ,Earth, Cell, Photo voltaic cell , Battery, Single pole single throw switch, Push-button switch, Energy meter, Alternator, Generator, D.C. Motor, Two-way switch, Fuse, Socket 2 pin, 3 pin, Aerial / Antenna, Voltmeter, Ammeter, Relay contacts, Transformer tapped secondary, Shielded wire,								5
3.	Circuit Design with special purpose diode: Zener Diode Applications: As reference voltage device, In constant current source with transistor, In over voltage protection, In clipping and clamping circuit, Clipper and clamper circuit, Dc power supply drawing, The Laser Diode, Optoisolator, Tunnel Diode, Tunnel Diode Oscillator, Varactor Diode, Application of Varactor Diode , Shockley Diode Testing of components using millimeter or oscilloscope								5

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4.	Circuit Design with Electronics components: Construction and symbols of IGBT, SCR, TRIAC, DIAC, Design of Instrumentation amplifier, Applications of IGBT, SCR, TRIAC, and DIAC.	5
5.	PCB Design guidelines: Introduction, Layout, Layout scale, Layout approach, Layout Procedure, PCB Types and Sizes: Single sided PCB and Double sided PCB.	4
6.	Design of oscillator and Amplifier: Oscillator design: Crystal Oscillator, Hartley oscillator, RC Phase Shift Oscillator, Colpitts Oscillators, Wien Bridge Oscillator. Amplifier: Class A Amplifier, Class B Amplifier, Class AB Amplifier, Class C Amplifier	4
TOTAL		28

List of Experiments:

Total 9 experiments: Any 4 experiment from 1 to 7 should be drawn on Sheets. 10th Experiment is compulsory. Any 3 experiments from 11 to 15. Any One experiment from 8 and 9.

1. Draw the symbols of various electronics and Electrical components: D.C., A.C. Positive. Negative. Single Phase A.C. 50 Hz. Three Phase A.C., 50 Hz. A.C. / D.C. 3-Phase line, Earth, Cell, Battery, Single pole single throw switch, Push-button switch, Energy meter, Alternator, Generator, D.C. Motor.
2. Draw the symbols of various electronics and Electrical components: A.C. Motor Single phase, 3-phase squirrel cage motor, 3-phase slip ring motor. Capacitor: Fixed, variable, Electrolytic Capacitor, Two-way switch, Fuse, Socket 2 pin, 3 pin, Aerial / Antenna, Voltmeter, Ammeter, Ohm Meter, Watt Meter, Lamp, Electric bell, Buzzer, Connections: star, Delta, Choke, Transformers, Resistor : Fixed, Resistor: variable, Diode, Auto transformer.
3. Draw the symbols of various electronics and Electrical components: Zener diode, Schottky diode, SCR, TRIAC, PNP transistor, NPN transistor, FET N-channel, FET P-channel, Unijunction transistor.
4. Draw inverter will once again connect the load, which are connected to its output to the main supply.
5. Draw Lamp dimmer cum universal speed controller circuit.
6. Draw circuit connections of a 12V, 1A regulated power supply using 7812.
7. Draw circuit of Stepped voltage stabilizer-manual.
8. Testing or measuring various electronics/electrical components using millimeter or oscilloscope.
9. Soldering techniques.
10. Design circuit with connection of resistors, capacitor in series and parallel. Simulate and measure voltages at each point. **(Compulsory)**
11. Design circuit drawing using Diode and special purpose diode and simulate it.
12. Design Simulate Battery and LEDs connected with switch measure voltages at each point
13. Design and simulate logic for controlling LED by using two switches.
14. Design and simulate Inverting and non-inverting amplifier
15. Design and simulate DC power supply and measure the voltages.

Text Books:

1. V.R. Deo, "Electronic Components & Application", Ane Books Pvt, Ltd.
2. Dwivedi and Tripathi, "Fundamentals of Electrical Engineering", Wiley Publication

Reference Books:

1. Paul Horowitz, Hill, "The art of Electronics", Cambridge University Press.
2. Floyd, "Electronics Devices", Pearson Publications.
3. O.N. Pandey, "Electronics Engineering", Ane Books Pvt, Ltd.

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| <ol style="list-style-type: none">4. Daniel Shanefield, "Industrial Electronics", Noyes Publications.5. Ganesh Babu and Suseela, "Linear Integrated circuit", Scintech Publications.6. J.R. Cogdell, "Foundation of Electronics", Prentice Hall.7. B.D.Shinde and Gitapathi, "Electronics and instrument System Design", Center of Technical Coordination. |
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E-Resources:

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| <ol style="list-style-type: none">1. Electrical and Electronics Symbols and Meanings:
https://www.edrawmax.com/article/electrical-and-electronic-symbols.html2. Electronics Tutorials:
https://www.electronics-tutorials.ws/resources/basic-schematic-symbols.html3. Electronics symbols: https://www.electronicshub.org/symbols/ |
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